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Roll No

EE-7004(4)/EX-7004(3)-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

SCADA Systems and Applications

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. What is SCADA system? Explain the working of SCADA system for monitoring application.
2.
 - a) Explain the programming language structure of PLC.
 - b) Explain in brief intelligent electronic devices in SCADA system.
3.
 - a) What is the basic components of SCADA? Explain each of the components.
 - b) Explain about the SCADA server system.
4.
 - a) Why communication technology is essential in SCADA system? Justify.
 - b) Explain in detail open standard communication protocols.
5.
 - a) Discuss the method of automatic substation control in interconnected power system.
 - b) Enlist various system operating states in power system. How system security feature may introduces? Explain.

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6.
 - a) Write short notes on System security in interconnected power system.
 - b) Write short notes on State estimation in interconnected power system.
7.
 - a) Explain the SCADA Energy management system.
 - b) How SCADA system help in transmission and distribution operation?
8. Write short notes on:
 - a) Fiber optics
 - b) Remote Terminal Unit
